ISBAT UNIVERSITY
A CHARTERED UNIVERSITY

DBMS – MYSQL

1) Database and Table Creation

1. Create a database called **dbisbat_enroll.**
2. Inside it, create a table **Student** with fields to hold:
    - **student_id** (Primary Key, AUTO_INCREMENT)
    - **first_name**
    - **last_name**
    - **class_code**
    - **registration_status** (Registered / Yet to Register)
    - **admission_date** (DATE)
    - **expertise** (Elective subjects/courses)
    - **tuition_fee** (DECIMAL)
    - **academic_level** (Junior / Mid / Senior)

3. TABLE VIEW DESIGN

student_id	first_name	last_name	class_code	registration_status	admission_date
1001	BADRU	NDAWULA KAYONDO	DSES24DA	Yet to Register	2024-08-15
1002	BYARUHANGA	FRANCIS	DSES24DA	Yet to Register	2024-08-16
1003	ABISHA	D'SOUZA	DSES24DA	Yet to Register	2024-08-16
1004	FAWADH	MOHAMMED	DSES24DA	Yet to Register	2024-08-17
1005	FRANCIS	LORO JOSEPH SOKIRI	DSES24DA	Registered	2024-08-10
1006	KABUYE	ISAAC LUYOMBO	DSES24DA	Registered	2024-08-11
1007	JIMMY JAMES	KABUYE	DSES24DA	Yet to Register	2024-08-18
1008	KADALA	HISHAM	DSES24DA	Registered	2024-08-12
1009	KEMIGISHA	ESTHER	DSES24DA	Yet to Register	2024-08-19
1010	KIGAANIRA	FAISAL	DSES24DA	Yet to Register	2024-08-19
1011	MALAGALA	DAVID	DSES24DA	Yet to Register	2024-08-20
1012	NALUKWATA	SHAMIRAH	DSES24DA	Registered	2024-08-13
1013	NAMPIINA	ESTHER	DSES24DA	Yet to Register	2024-08-21

specialization	tuition_fee	academic_level
DBMS, MySQL	2300.00	Junior
Python, DBMS	2400.00	Junior
Data Analysis, SQL	2200.00	Junior
Networks, DBMS	2450.00	Junior
DBMS, Java	2700.00	Mid
DBMS, Python	2550.00	Mid
Web, DBMS	2100.00	Junior
Java, DBMS	2650.00	Mid
DBMS, HTML/CSS	2000.00	Junior
DBMS, JavaScript	2800.00	Mid
DBMS, Linux	2350.00	Junior
DBMS, Data Viz	2900.00	Mid
DBMS, JavaScript	2600.00	Mid

2) Table Modifications

1. Change the **auto-increment starting value** for student_id to begin at **1001**.
2. Add a new column called **performance_rating** (INT).
3. Rename the column **expertise** to **specialization**.

3) Data Insertion

1. Insert records for the **13 students listed below.**
2. Add at least **five additional students** with recent admission dates (in **2024/2025**) for testing.

4) Data Retrieval

1. Display **all** student records.
2. Retrieve only **first_name, last_name, class_code** of all students.
3. Display **Full Name** and **tuition_fee** for students in class **DSES24DA**.
4. Find all students whose **specialization** includes **"JavaScript"**.
5. Retrieve names of students admitted **after 1st January 2022**.

5) Data Update

1. Promote all students to **Senior** level if their **admission_date** is more than **3 years** ago.
2. Change the **class_code** for **KABUYE JIMMY JAMES** to a different class (e.g., **DSES24DB**).

6) String Operations

1. Display **Full Names** by concatenating **first_name** and **last_name**.
2. List all students whose **first_name** contains the letter **'a'**.
3. Find students whose **last_name** starts with the letter **'K'**.

7) Tuition Analysis

1. Find the **average tuition fee** for students in class **DSES24DA**.
2. Find the **highest tuition fee** among all students.
3. List all students whose **tuition_fee** is **greater than 2,500**.

8) Data Deletion and Truncation

1. Delete a student record based on their **student_id**.
2. **Truncate** the table Student (remove all records but keep structure).